

Maksimum punte: <i>Maximum marks:</i>	16	Volpunte: <i>Full marks:</i>	15
Duur van vraestel: <i>Duration of paper:</i>	30 min + 10 min to complete OCR form <i>30 min + 10 min om OKH vorm te voltooi</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>	6 + OCR		

Important

- The examination regulations of the University of Pretoria apply.
- All questions must be answered on the Optical Character Recognition (OCR) form.
- Read the instructions on the OCR form carefully.
- Answers must include units, and must be written as rational numbers.

Dosent(e) / <i>Lecturer(s):</i>	D le Roux, D Ramotsoela, X Ye
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Do step 0 in the first 5 minutes of the session. (0-5 min)

Do step 1 in the first 30 minutes of the test. (5-35 min)

Do step 2 within the following 10 minutes. (35-45 min)

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form

STEP 1 (0 min to 30 min): Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (30 min to 40 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2018EBN111C3**
- Use a mechanical pencil with 0.5mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops.

- Complex numbers:

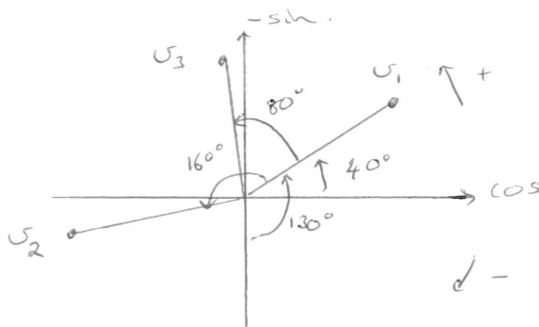
Rectangular form: always place the "j" **at the start** of the imaginary part of the complex number.

Polar form: always add the "°" (degree sign) after the angle, and make a clear angle sign.

Examples

01	-	1	.	4	+	j	8			μ	A						
02	3	4	0	∠	4	5	°			V							
03	-	j	5	4						k	Ω						
04	1	∠	2	0	°												
05	5	0	∠	-	1	3	0	°		m	A						

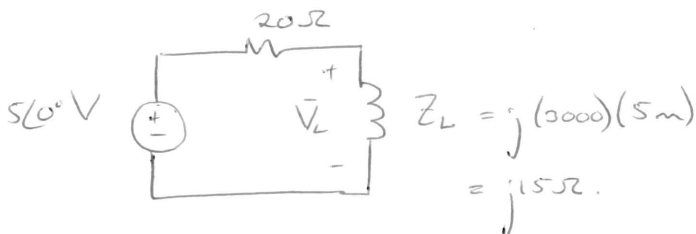
Questions 01-02 [2]		Vraag 01-02 [2]	
The following sinusoid is given: $v_1 = 16 \sin(\omega t + 130^\circ) \text{ V}$ It is known that - v_2 leads v_1 by 160°. - v_1 lags v_3 by 80°.		Die volgende sinusied word gegee: $v_1 = 16 \sin(\omega t + 130^\circ) \text{ V}$ Dit is bekend dat: - v_2 loop voor v_1 met 160°. - v_1 loop na v_3 met 80°.	
01	What is the phase angle ϕ_d between v_2 and v_3 ? ($0 \leq \phi_d \leq 180^\circ$)	01	Wat is die fasehoek ϕ_d tussen v_2 en v_3 ? ($0 \leq \phi_d \leq 180^\circ$)
02	Choose the correct statement: 1 v_2 leads v_3 2 v_3 leads v_2 (Write 1 if statement 1 is correct, and 2 if statement 2 is correct.)	02	Kies die korrekte stelling: 1 v_2 loop voor v_3 2 v_3 loop voor v_2 (Skryf 1 indien stelling 1 reg is, en 2 indien stelling 2 reg is.)



$$\phi_d = 80^\circ \quad \checkmark$$

1 $\rightarrow v_2$ leads v_3 . $\checkmark$

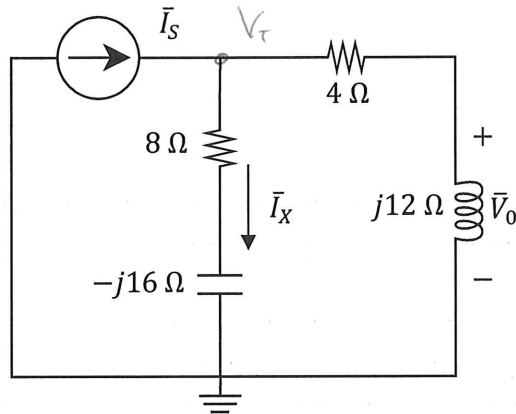
Question 03 [2]		Vraag 03 [2]	
The following three elements are all connected in series: a $5 \cos(3000t) \text{ V}$ voltage source, a 20Ω resistor, and a 5 mF inductor. What is the voltage across the inductor? (Give the answer in polar format . Assume the positive form of Ohm's Law.)		Die volgende drie elemente word in series gekoppel: 'n $5 \cos(3000t) \text{ V}$ spanningsbron, 'n 20Ω resistor, en 'n 5 mH induktor. Wat is die spanning oor die induktor? (Give the answer in polêre formaat . Aanvaar die positiewe vorm van Ohm se Wet.)	
03	$\bar{V}_L$	03	$\bar{V}_L$



V-division:

$$\bar{V}_L = \frac{5 \angle 0^\circ \times j15}{20 + j15} = 3 \angle 53.13^\circ \text{ V} \quad \checkmark$$

Questions 04-05 [4]		Vraag 04-05 [4]	
Given: $\bar{I}_x = 5 \text{ A}$. Determine $\bar{V}_0$ and $\bar{I}_S$.		Gegee: $\bar{I}_x = 5 \text{ A}$. Bepaal $\bar{V}_0$ en $\bar{I}_S$.	
04	$\bar{V}_0$ (Rectangular form)	04	$\bar{V}_0$ (Rechthoekige vorm)
05	$\bar{I}_S$ (Rectangular form)	05	$\bar{I}_S$ (Rechthoekige vorm)



$$* \quad V_T = I_x \times (8 - j16) = 40 - j80 \text{ V}$$

$$V\text{-division} \rightarrow V_0 = \frac{V_T \times j12}{4 + j12} = \underline{60 - j60 \text{ V}} \quad \checkmark$$

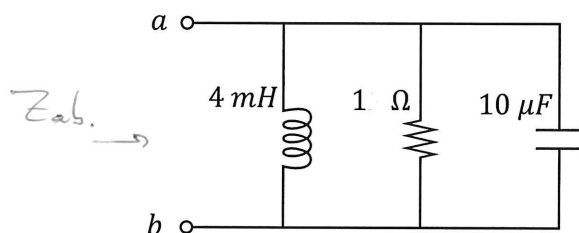
* I-division:

$$I_x = \frac{I_S \times (4 + j12)}{(4 + j12) + (8 - j16)}$$

$$5 = \frac{I_S (4 + j12)}{12 - j4}$$

$$I_S = \frac{5(12 - j4)}{4 + j12} = \underline{0 - j5 \text{ A}} \quad \checkmark$$

Question 06 [2]		Vraag 06 [2]	
06	For what value of ω will Z_{ab} be equal to: $Z_{ab} = 1 + j0 \Omega$ (Enter the unit as: $\text{rad} \div \text{s}$ on the OCR form.)	06	Vir watter waarde van ω sal Z_{ab} gelyk wees aan: $Z_{ab} = 1 + j0 \Omega$ (Skryf die eenheid as: $\text{rad} \div \text{s}$ op die OKH vorm.)



$$Z_c = \frac{1}{j\omega C} \quad Z_L = j\omega L$$

$$1 + j0 = \left(\frac{1}{1} + \frac{1}{j\omega L} + \frac{1}{\frac{1}{j\omega C}} \right)^{-1}$$

$$\frac{1}{1 + j0} = \frac{1}{1} + \frac{1}{j\omega L} + j\omega C$$

$$\frac{1}{j\omega L} + j\omega C = 0$$

$$\frac{1}{j\omega L} = -j\omega C$$

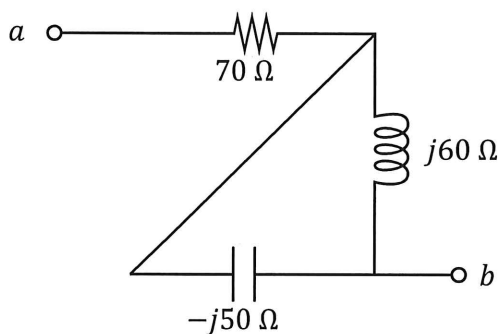
$$\frac{1}{LC} = -j^2 \omega^2$$

$$\omega = \frac{1}{\sqrt{LC}}$$

$$= \frac{1}{\sqrt{4\text{m} \times 10\mu\text{F}}}$$

$$= 5000 \text{ rad/s}$$

Question 07 [2]		Vraag 07 [2]	
07	Calculate Z_{ab} . (Rectangular format)	07	Bereken Z_{ab} . (Reghoekige formaat)



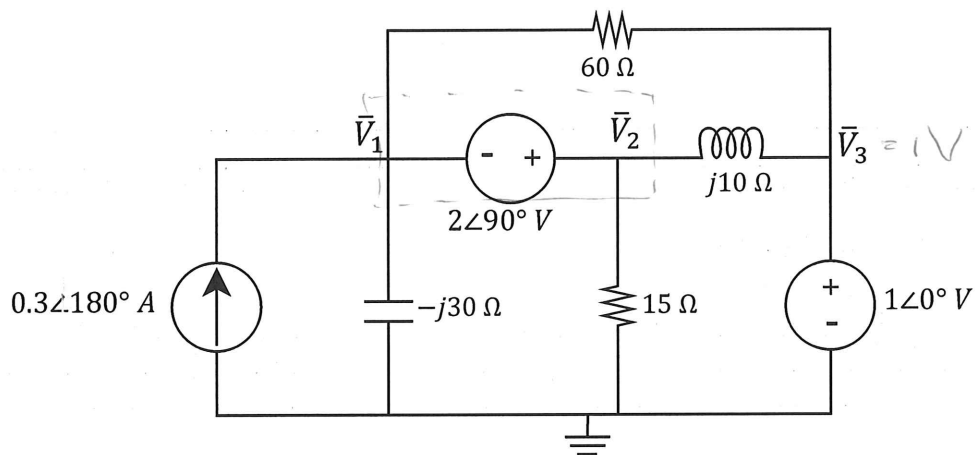
$$Z_{ab} = 70 + (-j50) \parallel (j60)$$

$$= 70 + \frac{(-j50)(j60)}{j60 - j50}$$

$$= 70 + \frac{-j^2 3000}{j10}$$

$$= 70 - j300 \Omega$$

Question 08-11 [4]				Vraag 09-12 [4]			
Use nodal analysis to set up two equations of the following form:				Use nodal analysis to set up two equations of the following form:			
$a\bar{V}_1 + b\bar{V}_2 = 6 - j17$				$a\bar{V}_1 + b\bar{V}_2 = 6 - j17$			
$c\bar{V}_1 + d\bar{V}_2 = j2$				$c\bar{V}_1 + d\bar{V}_2 = j2$			
Give the coefficients in rectangular form .				Give the coefficients in rectangular form .			
08	a	09	b	08	a	09	b
10	c	11	d	10	c	11	d



* Supernode: $V_2 - V_1 = 2\angle 90^\circ$

$\therefore -V_1 + V_2 = j2$

* KCL: $-0.3 = \frac{V_1}{-j30} + \frac{V_1 - 1}{60} + \frac{V_2 - 1}{j10} + \frac{V_2}{15}$

($\times j60$) $-j18 = -2V_1 + jV_1 - j + 6V_2 - 6 + j4V_2$

$6 - j17 = V_1(-2 + j) + V_2(6 + j4)$

PRACTICE PAGE / OEFENBLAD

Assessment ID

2	0	1	8	E	B	N	1	1	1	C	3		
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Remember:

Write a complex number in:

- rectangular form as $a \pm j b$ where j is at the **start** of the imaginary number.
- polar form as $a \angle \theta^\circ$ where the angle sign $\angle$ is clear, and you add the degree unit after the angle).

(See examples on page 1.)

	Answer	Unit
01	$\phi_d = 80^\circ$	
02	Statement #: 1	No unit required
03	$\bar{V}_L = 3 \angle 53.13^\circ$ (polar)	V
04	$\bar{V}_0 = 60 - j60$ (rectangular)	V
05	$\bar{I}_s = 0 - j5$ (rectangular)	A
06	$\omega = 5000$	rad ÷ s
07	$Z_{ab} = 70 - j300$ (rectangular)	Ω
08	$a = -2 + j$ (rectangular)	
09	$b = 6 + j4$ (rectangular)	
10	$c = -1$ (rectangular)	
11	$d = 1$ (rectangular)	